Higher National Diploma in Engineering (Mechanical/Electrical)
4th Year 1st semester Examination- 2019
ME 3227 – INDUSTRIAL ENGINEERING MANAGEMENT

Instructions for Candidates:
Answer **any 5** questions.

No. of questions : 06
No. of pages : 04
Time : 03 hours

01)

a) What is the use of OR? (02 marks)

b) State 3 advantages of using a model comparing to the actual situation. (03 marks)

c) A company produces two products X and Y, each of which requires three types of processing. The length of time for processing each unit and the profit per unit are given in the following table.

	Product X (hr/unit)	Product Y (hr/unit)	Available capacity per day (hr)
Process I	12	12	840
Process II	3	6	300
Process III	8	4	480
Profit per unit (Rs.)	5	7	

i) Write the objective function and all constraints. (05 marks)

ii) By using the graphical method find the number of units of each product that the company should be produced per day in order to maximize profit. (10 marks)

02)

A) a) Describe followings with suitable examples. (05 marks)

i) Fixed cost ii) Variable cost

b) The details for cost of production in a bakery is as follows.

Rent of the bakery premises	Rs. 10,000.00
Cost of flour for 100 loaves of bread	Rs. 2,000.00
Cost of labour for 100 loaves of bread	Rs. 500.00
Other fixed costs	Rs. 2,000.00

If the selling price of one unit is Rs. 50.00 find the number of loaves of bread to be produced in order to meet the break - even point. (05 marks)

02)

B) a) Describe the followings in brief. (05 marks)

i) Law of diminishing marginal utility
ii) Determination of market price

b) Discuss advantages and disadvantages of Perfect Competition market structure in the point of view of a consumer. (05 marks)

03)

a) Briefly describe the main categories of cash flows incurred in analyzing of engineering Projects. (04 marks)

b) A company has two projects under consideration for which the following cash flow forecasts have been made.

	Project 01 (Rs.)	**Project 02 (Rs.)**
Cash inflow at end of the year 1	16,000.00	20,000.00
Cash inflow at end of the year 2	40,000.00	60,000.00
Cash inflow at end of the year 3	80,000.00	40,000.00
Cash inflow at end of the year 4	60,000.00	40,000.00

Project 01 costs Rs. 140,000.00 and project 02 costs Rs. 110,000.00 as initial cost, which has to be paid immediately.

i) Calculate the present value of the cash inflows of the project at 10% discount rate. (04 marks)

ii) Which project should be undertaken by the company considering the following methods. Necessary calculations and explanation should be provided to verify your answer.

- Net present value (04 marks)
- Profitability index (04 marks)
- Payback period (04 marks)

04)

A)

a) What are the benefits of project management and planning? (04 marks)

b) State different types of organizational structures that are commonly observed in industrial environments. (04 marks)

c) Discuss the use of proper organizational structure in project management. (04 marks)

B)

a) Describe the term "Quality" with reference to some common quality dimensions. (04 marks)

b) Briefly explain any two quality control tools. (04 marks)

05)

a) Describe the role of Human Resource manager in an industrial organization with reference to key HR activities. (05 marks)

b) Discuss the ways that good HRM process can drive any organization towards the success. (05 marks)

c) What are the characteristics of a good leader? (05 marks)

d) How do you use leadership in order to motivate your followers in the workplace as an Engineering manager? (05 marks)

06)

a) An electronics manufacturer Textech, assembles model AT7S handheld calculators at his plant. The tasks that must be performed on each calculator are shown below. The parts used in this assembly line are supplied by materials-handling personnel to parts bins used in each task. The assemblies are moved along by belt conveyors between workstations. Textech wants this assembly line to produce 54 calculators per hour and the available time per hour is 54 minutes.

Task	Description	Predecessor	Time (minutes)
A	Place circuit frame on jig.		0.18
B	Place circuit no.01 into frame.	A	0.12
C	Place circuit no.02 into frame.	A	0.32
D	Place circuit no.03 into frame.	A	0.45
E	Attach circuits to frame.	B,C,D	0.51
F	Solder circuit connections to central circuit control	E	0.55
G	Place circuit assembly to calculator inner frame	F	0.38
H	Attach circuit assembly to calculator inner frame	G	0.42
I	Place and attach display to inner frame	H	0.30
J	Place and attach keyboard to inner frame	I	0.18
K	Place and attach top body of calculator to inner frame	J	0.36
L	Place and attach power assembly to inner frame	J	0.42
M	Place and attach bottom body of calculator to inner frame	K,L	0.48
N	Test circuit integrity	M	0.30
O	Place calculator and printed matter in box	N	0.39
		Total	5.36

i) Draw the network diagram. (04 marks)

ii) Compute the cycle time per calculator in minutes. (02 marks)

iii) Compute the minimum number of workstations. (02 marks)

iv) How would you combine the tasks into workstations in order to maintain the cycle time as well as to minimize idle time? (04 marks)

v) What is the efficiency? (02 marks)

b) Discuss the suitability of product layout for a manufacturing process providing examples and considering its both positive and negative impacts. (06 marks)